ARTICLE

Oral microbiota signatures in post-traumatic stress disorder (PTSD) veterans

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Post-traumatic stress disorder (PTSD) represents a global public health concern, affecting about 1 in 20 individuals. The symptoms of PTSD include intrusiveness (involuntary nightmares or flashbacks), avoidance of traumatic memories, negative alterations in cognition and mood (such as negative beliefs about oneself or social detachment), increased arousal and reactivity with irritable reckless behavior, concentration problems, and sleep disturbances. PTSD is also highly comorbid with anxiety, depression, and substance abuse. To advance the field from subjective, self-reported psychological measurements to objective molecular biomarkers while considering environmental influences, we examined a unique cohort of Israeli veterans who participated in the 1982 Lebanon war. Non-invasive oral 16S RNA sequencing was correlated with psychological phenotyping. Thus, a microbiota signature (i.e., decreased levels of the bacteria *sp_HMT_914, 332 and 871* and *Noxia*) was correlated with PTSD severity, as exemplified by intrusiveness, arousal, and reactivity, as well as additional psychopathological symptoms, including anxiety, hostility, memory difficulties, and idiopathic pain. In contrast, education duration correlated with significantly increased levels of *sp_HMT_871* and decreased levels of *Bacteroidetes* and *Firmicutes*, and presented an inverted correlation with adverse psychopathological measures. Air pollution was positively correlated with PTSD symptoms, psychopathological symptoms, and microbiota composition. Arousal and reactivity symptoms were correlated with reductions in transaldolase, an enzyme controlling a major cellular energy pathway, that potentially accelerates aging. In conclusion, the newly discovered bacterial signature, whether an outcome or a consequence of PTSD, could allow for objective soldier deployment and stratification according to decreases in *sp_HMT_914, 332, 871*, and *Noxia* levels, coupled with increases in *Bacteroidetes* levels. These findings also raise the possibility of microbiota pathway-related non-intrusive treatments for PTSD.

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INTRODUCTION

Despite advances in psychological diagnosis, post-traumatic stress disorder (PTSD), while being highly prevalent [1, 2], does not present definitive and objective biomarkers. Given how extensive stress exposure can damage individuals who are inherently more susceptible to stress, a wrong subjective psychological assessment could result in additional patient and societal damage [3]. War presents extreme stress-inducing situations, requiring that special attention be paid to soldier deployment, veteran benefits, and societal impacts. Considering the heritable nature of the disorder, the last decade was characterized by progress in pinpointing genetic biomarkers of PTSD based on genome-wide association studies. For example, a recent study on >250,000 veterans identified major genes as potentially causal or strongly related to PTSD [4, 5]. Accumulating somatic mutations throughout life may also change the course of PTSD, affecting neuro-immune interactions [6], with genetic and environmental factors further regulating the microbiota signature. For instance, *PLEKHM1*, a risk gene for PTSD [4], regulates *Salmonella* infection [7]. Following a similar rationale, our (and other) previous studies tied gut microbiota signatures to stressful psychological conditions, including autism and other intellectual disabilities [8–10], anxiety, and depression [11]. Similarly, oral microbiota composition has been tied to adolescent anxiety and depression symptoms [12], and schizophrenia [13]. Lastly, there is an excellent comprehensive study comparing gut microbiota in male cynomolgus macaques (*Macaca fascicularis*) including a chronic unpredictable mild stress group, or an unstressed control group. Microbe composition of paired mucosal and luminal samples from cecum, ascending, transverse, and descending colons were determined by 16S

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ribosomal RNA sequencing. Differences in mucosal and luminal microbial composition between the two groups were characterized by Firmicutes and Bacteroidetes at the phylum level, as well as Prevotellaceae and Lachnospiraceae at the family level, correlated with the depressive-like behavioral phenotype. Furthermore, significantly different microbiome community functions were discovered, mainly involved in carbohydrate and energy metabolism [14]. However, there is still a paucity of literature addressing gut microbiota and PTSD in humans, although an in-depth recent review comprehensively addressed the gut microbiome and mental health [15], and one particular study showed no significant general change, while a reduction in three bacterial phyla was correlated with higher PTSD severity [16].

To directly associate psychopathologies as well as protective measures related to PTSD (e.g., education attainment) [17] to oral microbiota composition, we re-visited (2018–2020) our well-controlled cohort of Israeli veterans who participated in the 1982 Lebanon war [3] and discovered a microbiome signature of PTSD. We further revealed a correlation of environmental air pollution with PTSD symptoms and psychopathology. Lastly, oral microbiome metabolic activity showed reductions in transaldolase, an enzyme controlling a major cellular energy pathway, that potentially accelerates aging [18], in correlation with excessive arousal and reactivity.

METHODS

Study cohort

In this study, we examined (2018–2020) our well-controlled cohort of Israeli veterans [3], in which 189 veterans were included, 108 of whom (57%) suffered from combat stress-related reactions during battle, and 33 (17.5%) were diagnosed as currently suffering from PTSD. Supplementary Table S1a details the age, ethnicity, income level, years of education (education attainment), and religiosity of the cohort.

Air pollution

We collected average annual $PM_{2.5}$ values for each subject from 2019. For each subject, we obtained the zip code of his/her residence, from which the city and region could be obtained from https://postalcode.48shops.com/Default.aspx?ln=english and https://isr.postcodebase.com/all. The official https://www.svivaaqm.net/ website reported average $PM_{2.5}$ values for 2019 from different air quality monitoring stations. For 2019, city-level annual average $PM_{2.5}$ data were calculated by averaging the annual average values of monitoring stations in each city. City-level annual average $PM_{2.5}$ data were then calculated by averaging the annual data from stations within each city. For cities without monitoring stations, region-level annual average $PM_{2.5}$ data were calculated by averaging the annual data from cities within each region. Each subject was matched to the average annual $PM_{2.5}$ data of the city in which he/she resided if the corresponding city data were available. Otherwise, the average annual $PM_{2.5}$ data of the region in which a subject resided was matched. In the end, a total of 186 subjects were assigned with either a $PM_{2.5}$ city or regional annual average, based on 2019 data, with three subjects not providing their city information. The mean annual exposure to $PM_{2.5}$ among the 186 subjects in 2019 was 19.4 $\mu g/m^3$. Maximum exposure was 22.6 $\mu g/m^3$ and minimum exposure was 17.1 $\mu g/m^3$. Supplementary Table S1b details $PM_{2.5}$ values, reflecting particulate matter with a diameter equal to or less than 2.5 microns, corresponding to the sizes of the key air pollutant, listed using zip code information.

Questionnaires

Study participants were subjected to a validated PTSD inventory questionnaire, adapted to the Diagnostic and Statistical Manual of Mental Disorders – V [19]. The PTSD inventory questionnaire [20] was combined with deep individual phenotyping, including eight additional questionnaires addressing demography, psychopathology, well-being, and physical health. These questionnaires addressed psychopathological symptomatology (Symptom Checklist-90-Revised), including but not limited to anxiety, hostility, and additional symptomatology, encompassing sleep, appetite disturbances, guilt feelings, as well as suicidal thoughts [21], social support [22], dyadic adjustment (marital support) [23], positive well-being [24, 25], satisfaction with life [26], post-traumatic growth (the capacity to grow out and above a traumatic event) (Supplementary Table S2) [27] and physical health [3]. For further details, see Supplementary materials.

Saliva sample collection

Following data collection, saliva samples were collected using the Oragene DNA (OG-500) vial collection kit (DNA Genotek, Ottawa, ON, Canada). Samples were coded and kept at room temperature according to the manufacturer's instructions. The study was reviewed and approved by the University of Haifa IRB (102\14). Ella Levert-Levitt and Professor Abraham Sagi-Schwartz completed NIH training on Protecting Human Research Participants (Certification Numbers: 1313252, 197741). Professor Illana Gozes's and Professor Zahava Solomon's research was approved by the Helsinki Committee of Tel Aviv University. Each of the participants signed an informed consent form.

DNA preparation

Salivary DNA was purified from the 189 individuals listed above. DNA was prepared at The Center for Genomic Technologies, The Institute of Life Sciences, Hebrew University of Jerusalem, Israel. Briefly, the prepIT•L2P reagent for purification of genomic DNA (DNA Genotek) was used. DNA was extracted following a 50 °C incubation (2 h), a wash, and ethanol precipitation to yield pure DNA samples.

16S RNA sequencing and bioinformatics

DNA samples ($N = 189$) were sequenced by Macrogen Europe using the Herculase II Fusion DNA Polymerase Nextera XT Index Kit V2, on an Illumina platform. Paired-end raw sequences (read lengths of 301 bps) were imported into qiime2 2020.11 [28], de-noised using q2-dada2 [29], and the resulting sequence variants were classified using q2-feature-classifier [30], trained on the HOMD 16S rRNA RefSeq V15.22 database [31]. Pathway abundance was measured using PICRUSt2 2.3.0b0 [32]. Downstream analysis was performed in R 4.0.3, using the phyloseq package [33] and ANCOMBC [34] for differential abundance analysis. ComplexHeatmaps was used for heatmap creation [35]. For the full list of packages used and their versions, see session information in the code repository (BioProject ID PRJNA844909; website link: https://github.com/shepp-hub/oral_microbiome_ptsd).

For each taxonomic rank, taxa abundances were aggregated, with the exception of uncharacterized and low-abundance taxa, which were removed. A single extreme outlier sample was omitted from the analysis. Differential abundance comparisons were made by comparing aggregated taxa of Bacteria for each rank from phylum to species.

Statistical information

The false discovery rate was used for multiple comparison corrections, where applicable [36], and results with FDR < 0.05 were considered significant. Differential abundance analysis was performed using the ANCOMBC method for each taxonomic rank separately. In cases where the same taxon showed differential abundance in multiple comparisons, all relevant parameters were added as covariates to the ANCOMBC formula, to rule out possible confounders. Microbial alpha diversity comparisons were made with the Chao1 index, using the Wilcoxon test. The taxon relative abundance heatmap was normalized using taxon-wide *Z*-scoring and clustered using K-means, calculated from the Euclidean distance.

RESULTS

Psychological measures are highly correlated and influenced by air pollution

A high expected correlation among the different psychological measures was indicated by the matrix shown in Fig. 1 (descriptive statistics are provided in Supplementary Table S3). Our results showed that $PM_{2.5}$ exposure level, reflective of air pollution, was positively correlated with PTSD and elevated psychopathology measures, including intrusive severity, negative cognition, and mood and total severity, as well as psychopathology total severity, encompassing obsessive-compulsive problems, hostility, interpersonal sensitivity, anxiety, and depression. Furthermore, $PM_{2.5}$ was negatively associated with subjective well-being and life satisfaction (Fig. 1).

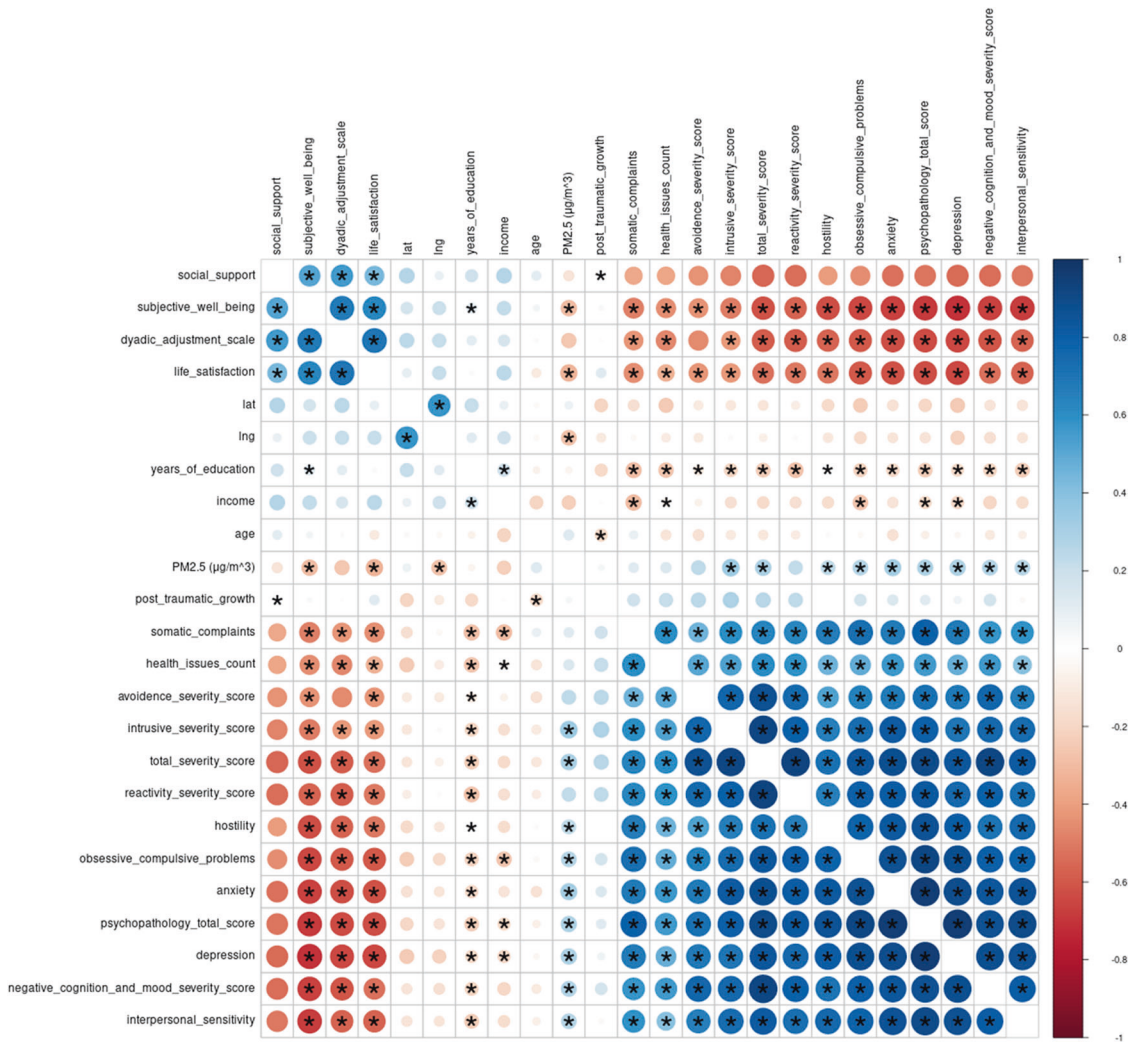


Fig. 1 Psychological measures are highly correlated and positively influenced by air pollution. Correlation matrix between pairs of parameters. Each circle represents a correlation between the row and column parameters, colored according to the trend of the correlation (color scale shown on the right) and sized according to the magnitude of the correlation. Significant correlations are marked with an asterisk (lng and lat are longitude and latitude geographic locations that are linked with air pollution, $PM_{2.5}$).

PTSD and psychopathological observations are strongly correlated with salivary microbiota diversity

Employing advanced bioinformatics tools, we discovered oral microbiota stratification paralleling the psychological measures, while contrasting with variables of well-being. Most pronouncedly, a reduction in oral microbiota diversity and abundance was associated with PTSD and psychopathology. Figure 2a and Supplementary Fig. S1 show that the most significant effects on oral microbiota composition were associated with smoking and alcohol abuse. In the 34 (17.7%) smokers in our tested population, decreases were noted in >25 bacterial species and increases were seen in ~20 strains (Supplementary Fig. S2a). In 11 (~6%) of our population who consumed alcohol daily, there were increases in ~40 bacterial species, and decreases in ~5 strains (Supplementary Fig. S2b).

Significant bacterial load changes were further observed in association with chronic idiopathic pain (levels of >20 species were reduced and of ~5 species were increased) and with additional psychopathological symptomatology, including sleep and appetite disturbances, guilt feelings, and suicidal thoughts (levels of 20 species were reduced and of ~5 species were increased). Similar associations were seen with arousal and reactivity symptom counts, including angry outbursts, hypervigilance, reckless and irritable behavior, exaggerated startle response, concentration problems, and sleep disturbance (levels of ~13 species were reduced and of ~8 species were increased) (Fig. 2a and Supplementary Fig. S1). PTSD measurements were calculated in two ways, either as a sum of scores or as a mean of scores, such as reactivity symptom counts and severity (score), respectively, for example. Both approaches revealed similar changes in overall bacterial load.

The table within Fig. 2a describing changes in bacterial diversity shows a significant reduction of microbiota diversity in association with recent weight gain ($N = 45$, 24% of the study cohort; Supplementary Fig. S3a, boxplot), weakness, and fatigue ($N = 60$, 31% of the study cohort; Supplementary Fig. S3b), heart problems ($N = 53$, 28% of the study cohort; Supplementary Fig. S3c), ulcer ($N = 12$, 6% of the study cohort; Supplementary Fig. S3d) and memory difficulties ($N = 60$, 31% of the study cohort; Fig. 2b). An almost significant reduction in bacterial diversity was also observed in individuals currently suffering from PTSD ($N = 33$, 17.5% of the tested population; Fig. 2c). Interestingly, most recent findings associate PTSD with reduced coronary micro-circulatory function and greater deterioration over time [37].

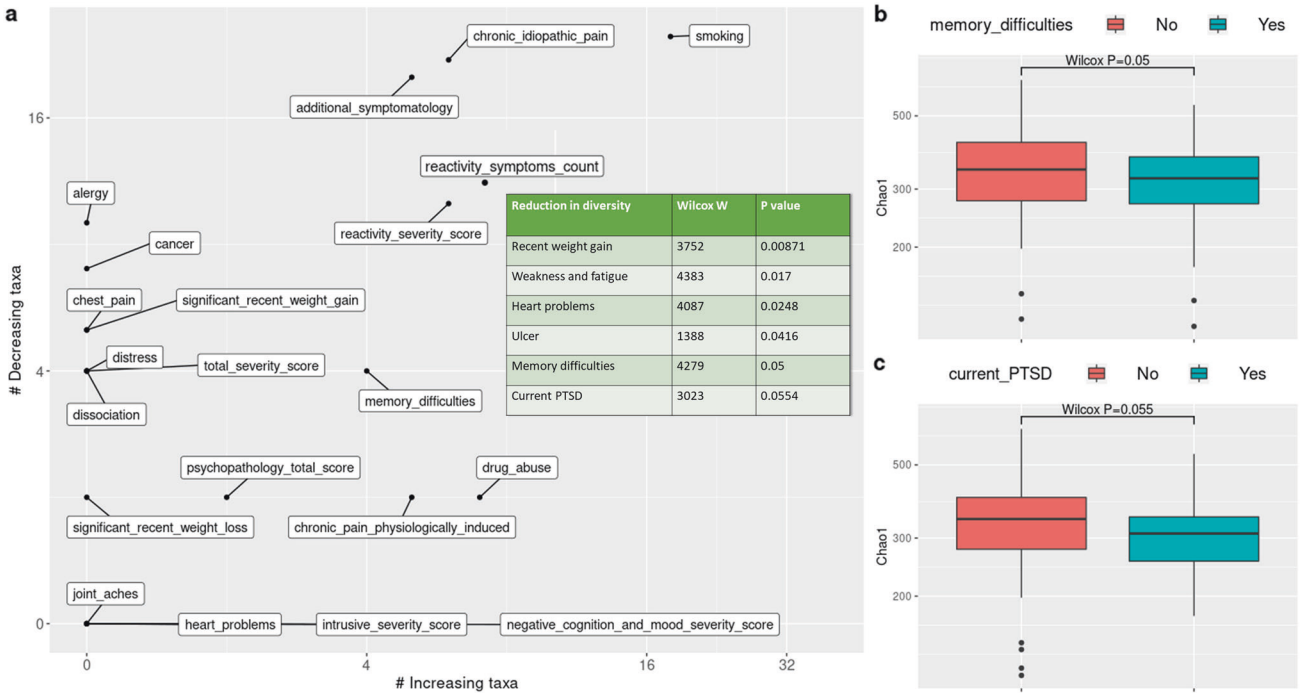


Fig. 2 Oral microbiota diversity associated with clinical and psychological issues. a A logarithmic scatter-plot of clinical and psychopathological parameters according to the number of increasing (*x*-axis) and decreasing (*y*-axis) microbial taxa abundance, when compared with unaffected controls. Table: clinical and psychopathological parameters that altered microbial diversity, along with test statistics and significance. **b**, **c** Boxplot comparisons of Chao1 diversity metrics for participants currently experiencing memory difficulties (**b**) or PTSD (**c**).

Individual oral microbiota composition stratifies PTSD

A heatmap comparing microbiota distribution among all 189 participants vs. psychological outcome measures and answers to well-being questionnaires, again showing alcohol abuse and smoking at the extremes, was generated (Fig. 3). Individuals with PTSD and corresponding high psychopathology measures in the tested cohort exhibited very similar bacterial distributions. This similarity extended to pain and recent weight loss but did not hold true when potentially protective measures, including years of education and life satisfaction, were considered. Specifically, microbiota changes associated with reactivity symptom counts revealed an opposite distribution to that obtained when addressing years of education, yet was similar to the distribution associated with memory difficulties.

Further use of volcano scatterplots that depict statistical significance (*P* value) versus magnitude of change (fold change) enabled quick visual identification of major changes in the microbiota. Graphs are presented in the alphabetic order of the specific questionnaire (Fig. 4). The results identified a microbiota signature for PTSD total severity score, which averages the results of the PTSD questionnaire. Signature bacterial changes included significant decreases in the levels of *sp_HMT_914*, *332* and *871*, and *Noxia*. When looking at PTSD total symptom counts, only *sp_HMT_914* levels decreased, while those of *Bacteroidetes* increased (Supplementary Fig. S4a). One of the four symptom groups of the PTSD score, namely reactivity symptom count (that includes symptoms of irritability, sleep disturbances, recklessness, hypervigilance, and concentration problems), was associated with the most robust changes in bacterial abundance (Figs. 2 and 4), and showed an overall similar result, as compared to the total severity score (Supplementary Fig. S4b). Thus, the microbiota signature associated with reactivity symptom count included a reduction in *Pasteri* levels and robust increases in the levels of *Bacteroidetes (Bacteroidia)*, smoking-linked *Veillonellales*, *Prevotella*, and *Firmicutes* (Fig. 4 vs. Supplementary Fig. S4b). Further assessing the levels of the PTSD-linked traits of negative cognition or mood and intrusive symptoms, the first showed a negative correlation with *Johnsonella* levels and the second with *sp._HMT_914* (Supplementary Fig. S4c, d). Significant decreases in *sp_HMT_914* levels were also shared with chronic idiopathic pain (Supplementary Fig. S4e), emphasizing the importance of holistic medical approaches.

An oral bacterial signature for memory difficulties and additional symptomatology measures

Memory difficulties (Fig. 4), were negatively correlated with *sp._HMT_332* and *Noxia* levels and positively correlated with *Bacteroidales* and *Bacteroidetes (Bacteroidia)* levels, pheno-copying what was seen with PTSD total score, and reactivity symptom count. High scores in additional symptomatology measures touching upon disturbances in appetite (i.e., lack of appetite on the one hand and over-indulgence in eating on the other, disturbed sleep patterns, as well as difficulties falling asleep, death-related thoughts, and guilt feelings) showed a dramatic decrease in *sp. HMT_332* levels.

Psychopathology total scores (Fig. 4), including individual measures of anxiety and hostility, indicated reduced amounts of *sp._HMT_914*, as characterized by several PTSD measures. Anxiety and hostility further correlated with decreases in *Pasteri*, and *Bacterium_HMT_871*, to a large degree mimicking the reactivity measures. A decrease in *sp. HMT_332* levels was associated with anxiety and was also linked with PTSD arousal and reactivity, memory difficulties, and additional symptomatology. A further reduction in the numbers of *Noxia* was correlated with the psychopathology total score and the individual measure of anxiety as part of the total PTSD severity score, reactivity severity count, and score and memory difficulties, but not in association with hostility. Anxiety and hostility further correlated with decreases in the levels of *Bacterium_HMT_871* and *Pasteri*, the first sharing a trend with PTSD severity score and reactivity measures, and the

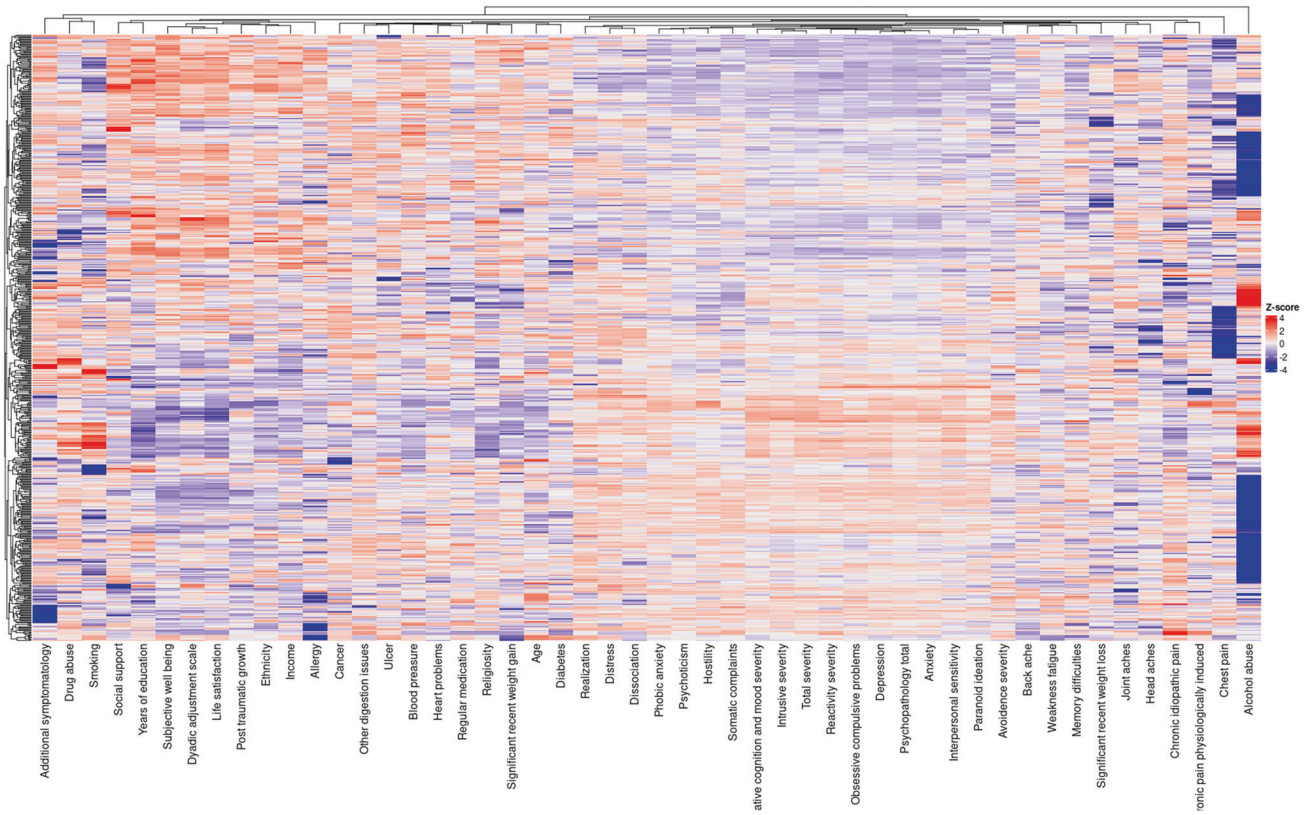


Fig. 3 Single taxon distribution analysis distinguishes affected and unaffected individuals. A clustered heatmap of *Z*-scored differential abundance statistics according to clinical and psychopathological parameters. Each column represents a comparison of results between affected and unaffected participants and each row represents a single taxon that showed significant differential abundance in at least one comparison.

latter with PTSD reactivity measures (Fig. 4 and Supplementary Fig. S4b). Increases in *Bacteroidetes* and *Firmicutes* levels were noted only in association with the psychopathology total score and anxiety (mimicking reactivity measures). Phobic anxiety correlated with decreases in the levels of *Noxia*, *sp._HMT_914*, and *Bacterium_HMT_871* (Supplementary Fig. S4f).

Opposite microbiota signatures are associated with years of education and psychopathological measures

As shown in the heatmap (Fig. 3), the microbiota profile associated with the years of education variable contrasted with that of the reactivity count measure (Fig. 4). This was specifically apparent in the volcano plot, where a dramatic increase in *Bacterium sp. HMT 871* levels is seen as a function of years of education (Fig. 4). This is in opposition to the decreased levels associated with reactivity count, anxiety, and hostility. At the same time, increased education was coupled with decreases in the levels of *Bacteroidetes* and *Firmicutes*, which showed the opposite trend as a function of reactivity count measure, psychopathology total score, and anxiety.

Specific microbiota signatures for social support, marital status, and blood pressure medications showing no association with psychopathology

Levels of social support, seen in the volcano plot (Supplementary Fig. S5a), displayed a negative correlation with *Pasteurellales*, *Peptostreptococcus*, *Fusobacterium*, and *stomatis* levels and positively correlated with levels of *Carnobacteriaceae*, *Granulicatella*, *adiacens*, *Gammaproteobacteria*, *parainfluenzae*, *Pasteurellaceae*, *Haemophilus*, *Bacilli*, *Fusobacteriaceae*, and *Kingella*. This pattern is completely different from the microbiome taxonomy found for individuals with PTSD and psychological symptomatology. Similarly, albeit with other bacterial changes, the dyadic adjustment scale assessing marital support was correlated with increases in microbiota groups not seen with any other measure (Supplementary Fig. S5b). No correlation in microbiota profile was observed with general medication use (Supplementary Fig. S5c). However, a correlation with the use of blood pressure medication was observed, also revealing a picture not seen with any other measure (Supplementary Fig. S5d), indicating specificity.

A unique microbiota signature for PTSD and related psychopathology

Focusing on changes in bacterial abundance, we discovered a unique oral microbiota signature for PTSD and related psychopathology, including 15 bacterial groups that showed significant correlations with at least two psychological measures and years of education (Fig. 5a). These species were enriched as a function of reactivity symptom count, as well as of anxiety symptomatology (Figs. 4 and 5a). In addition, memory difficulties shared a bacterial signature with other psychopathologies, such as PTSD and anxiety (Fig. 5a). The inverse profile patterns seen as a function of education and psychiatric symptomatology are intriguing findings, given that various variables of well-being assessed in the current study, such as satisfaction with life and subjective well-being, had no significant correlations with microbiome composition. On the other hand, levels of *Bacteroidia*, *Bacteroidetes*, and *Firmicutes* were positively correlated with obsessive-compulsive problems (Supplementary Fig. S6a), whereas interpersonal sensitivity (Supplementary Fig. S6b) and paranoid ideation (Supplementary Fig. S6c) were linked with increases in only *Bacteroidetes* and *Firmicutes* and a marked decrease in *Noxia* levels. Finally, the decreases in *Noxia* and *sp._HMT_914* levels were directly associated with psychoticism (Supplementary Fig. S6d), while increases in *Bacteroidetes*

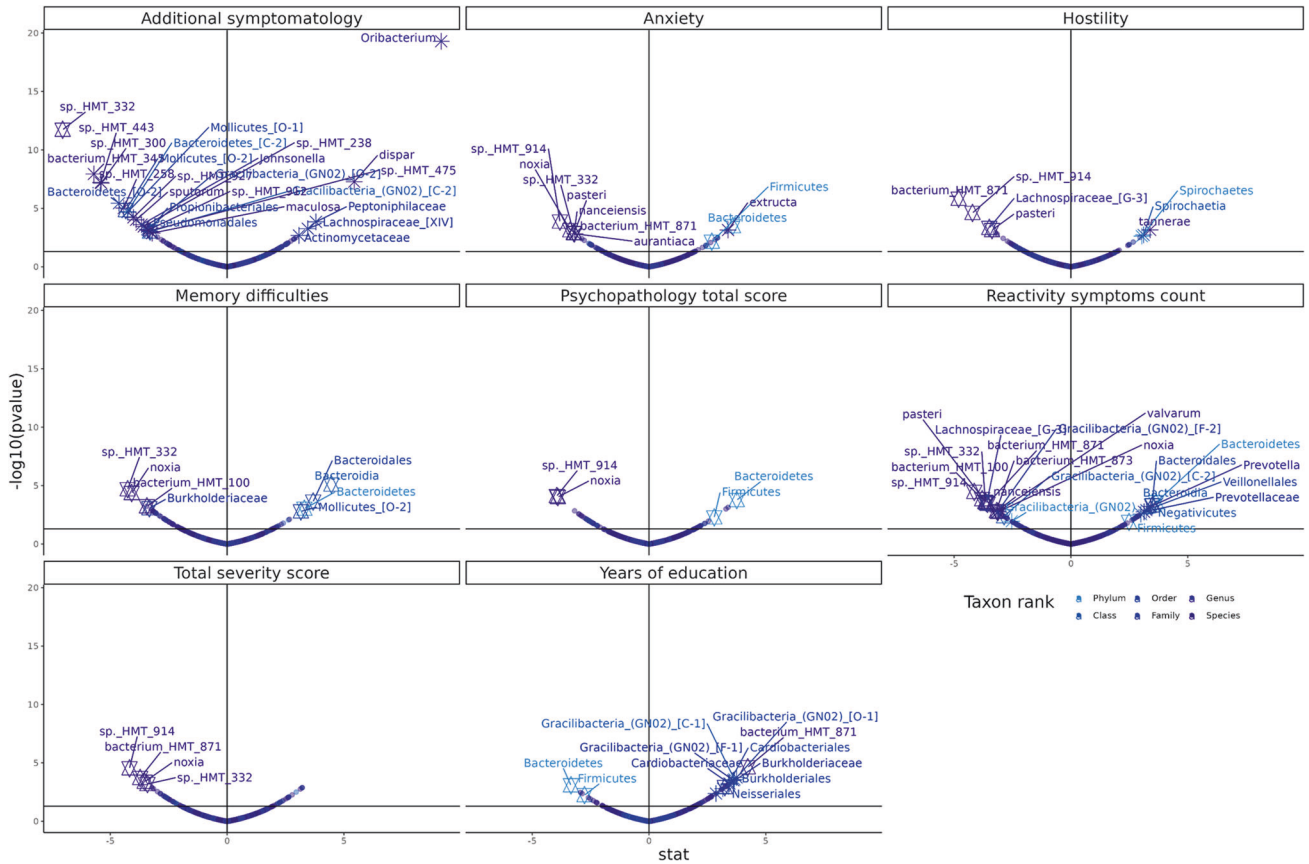


Fig. 4 The oral microbiota profile linked with exaggerated anxiety, arousal, and reactivity is opposed to the microbiota composition associated with education. Volcano plots visualizing differential abundances, with each representing a single comparison of affected and unaffected participants. Each taxon is represented by a single point, placed according to differential abundance statistics (*x*-axis) and significance (*y*-axis), with significantly differential abundances labeled with the name of the taxon. Recurring taxa are marked by ✡. Taxa are colored according to their phylogenetic rank.

and *Firmicutes* numbers were linked with depression (Supplementary Fig. S6e).

Verifications focusing on single bacterial groups showed, for example, a logarithmic reduction in *Bacteroidetes* vs. increases in *sp_HMT_871* levels as a function of years of education (Fig. 5b). This was in contrast to the increases of *Bacteroidetes* abundance and decreases in *Noxia* levels associated with memory difficulties (Supplementary Fig. S7). A dramatic decrease in *sp_HMT_871* levels was associated with additional symptomatology (Supplementary Fig. S7), while a reduction in *Noxia* numbers was also linked to memory difficulties and significant recent weight loss (Fig. 5c, d, respectively).

Connecting microbiota composition to metabolic activities and reactivity symptom count revealed highly significant differences in terms of enzymes and pathways associated with energy metabolism (Supplementary Fig. S8). These included transaldolase (TALDO1), a rate-limiting enzyme important for the balance of metabolites in the pentose-phosphate pathway, a major cellular energy regulatory pathway. Aberrant TALDO1 activity has been implicated in various autoimmune diseases and malignancies [38], possibly related to the observed bacterial changes associated with significant recent weight loss (Fig. 5d), and how the highly significant reduction in oral bacterial diversity was associated with recent significant weight gain (Fig. 2a, insert). Together, our results argue for an in-depth medical, sociological, and environmental assessment, as well as objective oral bacterial composition assessments for the prevention and diagnosis of PTSD, as well as the development of novel treatment modalities for the affliction.

DISCUSSION

To our knowledge, this is the first description of an oral microbiome signature among veterans with PTSD. In our cohort, approximately a third of the participants who suffered from PTSD did not receive any recognition from the Ministry of Defense, and as such were not entitled to receive rehabilitation, medical and therapeutic aid despite their longstanding daily coping with a debilitating mental condition. The newly discovered bacterial signatures described here, whether an outcome or a consequence of the condition, could potentially allow objective soldier deployment and patient stratification according to decreases in *sp_HMT_914, 332*, and *871* and *Noxia* levels coupled with increases in *Bacteroidetes* abundance, including microbiota pathway-related treatments for PTSD in a non-intrusive manner.

Regarding our observation of air pollution levels correlating with PTSD measures (Fig. 1), it is clear that healthy air also plays a major role in psychopathology outcomes. A study of >300,000 participants [39] found that the risks of nerves, anxiety, tension, or depression increased by 2.31 times per 10 μg/m^3 increase in $PM_{2.5}$, while the risk of major depression increased by 2.26 times per 10 μg/m^3 increase in $PM_{2.5}$. Another recent study similarly found that a 10 μg/m^3 reduction in $PM_{2.5}$ concentration was associated with a 4.14% drop in the depressive score [40], complementing our results here showing a relationship between $PM_{2.5}$ levels and PTSD.

Our microbiota signature findings showing reactivity symptom counts, psychopathology total score, and anxiety being associated with increases in the levels of *Firmicutes and Bacteroidetes* (Fig. 4) have been further corroborated by studies assessing murine fecal

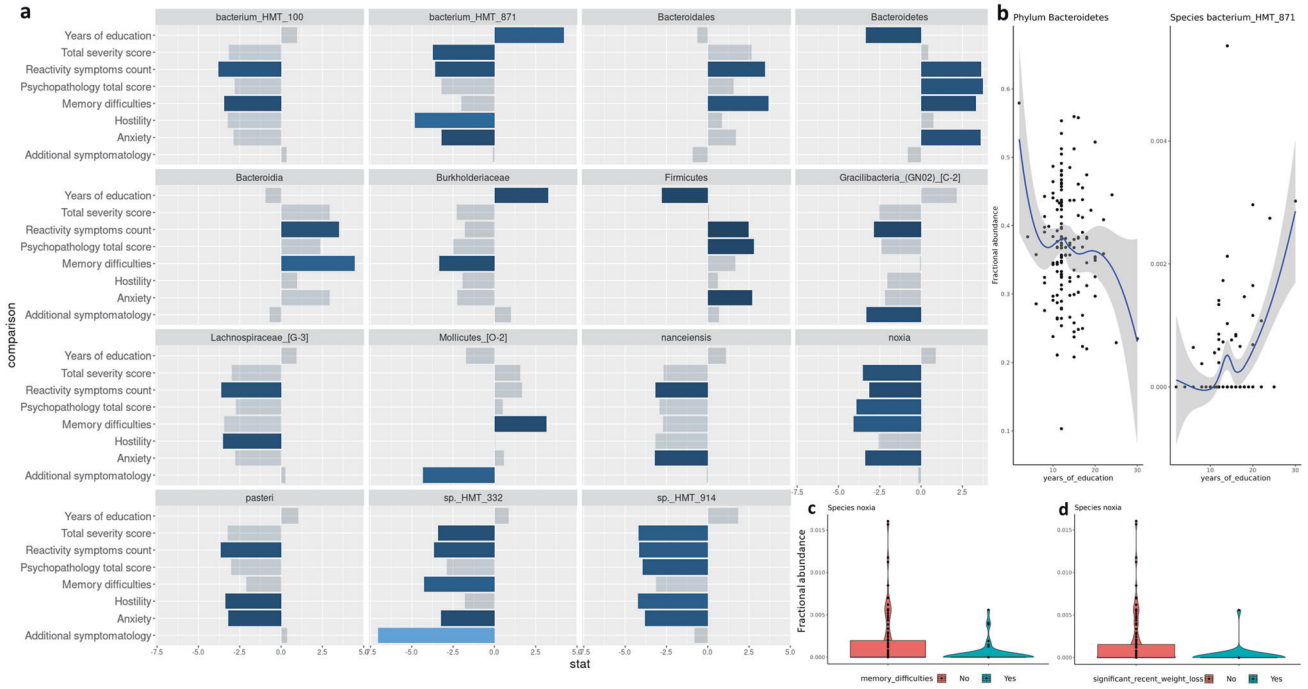


Fig. 5 Oral microbiota signatures in PTSD veterans. a Bar plots of all taxa with significantly differential abundances in multiple comparisons. Each panel represents a taxon, compared across different parameters (*y*-axis), with the trend and magnitude of differential abundances signified by the bar (*x*-axis) and colored according to significance. Opaque bars represent significantly differential abundance. **b** The fractional abundance of phylum *Bacteroidetes* and species *Bacterium HMT 871* (*x*-axis), compared to years of education (*y*-axis). Both correlations are statistically significant. **c** The fractional abundance of the species *Noxia* in participants experiencing memory difficulties. **d** The fractional abundance of the species *Noxia* in participants experiencing recent significant weight loss.

microbiota composition [41]. Most importantly, these findings are further corroborated by in-depth studies in male cynomolgus macaques (*Macaca fascicularis* [14]), indicating interesting evolutionary conservation of microbiota-behavior association, and strengthening the use of animal models.

It is also important to note that anxiety, highly manifested in schizophrenia, was recently addressed in studies that reported stepwise increases in salivary *Firmicutes* levels from healthy controls through clinical high-risk individuals to drug-naïve, first-episode schizophrenia patients [13]. Similarly, the diversity and relative abundances of oral bacterial communities were significantly altered in chronic insomnia patients but not in healthy individuals, with the phyla *Bacteroidetes* and *Gracilibacteria* being identified as major players [42]. Again, the levels of these bacteria changed in opposing directions in the present study when comparing reactivity (including the measuring of sleep problems) and years of education (Fig. 5). Increases in *Bacteroidetes* levels were also shown here as a function of reactivity, psychopathology, and anxiety, all including sleep disturbances (Fig. 5a). Addressing non-recurring bacterial changes, increases in *Prevotella* abundance was associated here with reactivity (Fig. 4), and independently associated with sleep characteristics [43], while the fecal *Prevotella*-to-*Bacteroides* ratio predicted body weight and fat loss success in a 24-week diet regimen [44], and mouse fecal *Bacteroides* levels were correlated with gait deficiencies [10]. Interestingly, *Bacteroides/Prevotella spp*. levels are regulated by pituitary adenylate cyclase-activating polypeptide (PACAP) [45], as well as by the leading autism and intellectual disability activity-dependent neuroprotective protein (ADNP) [9], regulating stress [8], PTSD [46], and learning and memory [43]. Furthermore, *Burkholderiaceae* levels exhibited a contrasting relation between memory difficulties (positive correlation) and years of education (negative correlation) (Fig. 5a), with proteins from these bacterial groups having been shown to boost immune memory [47]. Of note, *Noxia* level increases may be linked with reduced tooth cleaning [48], with *Selenomonas noxia* being an important periodontal pathogen, associated with gingivitis and periodontitis [49]. In turn, gingivitis [49] may be associated with the development of Alzheimer's disease [50], with PTSD potentially presenting a major risk factor for Alzheimer's disease onset and progression [6, 51].

Our findings highlight bacterial groups that have not been previously associated with PTSD or any other psychopathologies to date, further suggesting the usefulness of specific oral microbiota for diagnostics and potential treatment. Interestingly, a major change in oral bacterial diversity in association with alcohol abuse and smoking was discovered here (Fig. 2). Notably, the gut microbiota has been implicated in alcohol abuse, both from a diagnostic and a causative perspective, as well as in terms of curative characteristics [52]. Similarly, the oral microbiota composition has been observed to be influenced by cigarette smoking. For example, two nitrite-producing bacteria that can increase the acidity of the oral cavity, *Actinomyces* and *Veillonella*, and previously shown to be enriched in smokers [40], were identified here (Supplementary Fig. S2a). These findings might be considered as a validation of our methodology and results, while emphasizing changes in oral microbiota as an outcome of behavior, in this case, smoking.

In conclusion, our findings suggest that cleaner air (as reflected by reduced $PM_{2.5}$ concentration), coupling with higher education, provides resilience, characterized by a decrease in the abundance of selected strains, as compared to the effects on these bacteria as a function of pathological variables, such as PTSD (specifically, the arousal and reactivity dimensions), hostility and anxiety, as well as memory problems. As noted above, when connecting microbiota composition to metabolic activities and reactivity symptom counts, highly significant differences were discovered with enzymes and pathways associated with energy metabolism. Most important, similar findings were also obtained in male cynomolgus macaques with depressive-like behaviors [14], which are

highly comorbid with PTSD [53], thus validating and strengthening our findings.

Furthermore, nicotinamide mononucleotide (NAD), regulating energy metabolism, is a pivotal controller of healthy aging [18]. NAD supplementation corrects deficits in oxytocin, sociability, and anxiety in a mouse model of autism spectrum disorder [54]. In turn, microbiota changes are correlated with autistic behaviors [8, 55]. Notably, PTSD accelerates aging and may double the risk of mortality, as reviewed [56], further strengthening the potential use of NAD and related compounds. Collectively, our results raise the possibility for future objective individual diagnosis and microbiota-associated drug development, with the ultimate aim of personalized prevention and treatment of PTSD.

CODE AVAILABILITY

Repository code: BioProject ID PRJNA844909 (web site link: https://github.com/shepp-hub/oral_microbiome_ptsd).

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AUTHOR CONTRIBUTIONS

EL-L designed the study, contacted and interviewed the study participants, handled all the psychological questionnaires, collected DNA, summarized the results, and significantly contributed to paper writing. This study is in partial fulfillment of the PhD requirements of EL-L. GS performed all the bioinformatics, analyzed all data and provided all graphs, as well as edited the paper. SS helped in experimental design, DNA preparation for sequencing, and paper editing. NS provided bioinformatics expertise. JCKL and VOKL provided the air pollutant data and analyzed the correlation between air pollution and PTSD variables. MMH and SB provided microbiota expertise. ABY helped with the study cohort and funding. AS-S mentored EL-L and was instrumental in the design and execution of the study, focusing on the detailed psychological assessment. ZS provided the study cohort, connected the collaborating parties, and contributed to the psychological/sociological assessments. IG oversaw the study, provided biochemical, physiological and neuroscience expertise, integrated the data, and significantly contributed to paper writing.

COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

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